

CPU Architecture & Register Organization

Q. Explain the internal organization of a CPU. Describe the basic components including Registers, ALU, and the Control Unit.

📌 Definition to Remember

The **CPU (Central Processing Unit)** is the brain of the computer. Its internal architecture defines how it fetches, decodes, and executes instructions. It consists of three primary components interconnected by an internal bus:

Registers (storage), **ALU** (processing), and the **Control Unit** (management).

1. Registers

High-speed, temporary storage locations situated directly on the CPU chip.

- * **General Purpose Registers (GPRs):** Used by programmers to temporarily store data during arithmetic operations.
- * **Special Purpose Registers (SPRs):**
- * **Program Counter (PC):** Holds the address of the *next* instruction to be fetched.
- * **Instruction Register (IR):** Holds the *current* instruction being decoded/executed.
- * **Memory Address Register (MAR):** Holds the memory address for reading/writing.
- * **Memory Buffer Register (MBR/MDR):** Holds the actual data fetched from memory.
- * **Accumulator (ACC):** Stores intermediate mathematical results.

2. Arithmetic and Logic Unit (ALU)

The execution core where actual data processing happens.

- * **Arithmetic Operations:** Addition, Subtraction, Multiplication, Division.
- * **Logical Operations:** AND, OR, NOT, XOR, Shift, Compare.
- * Outputs results to the Accumulator or updates **Status Flags** (Zero flag, Carry flag).

3. Control Unit (CU)

The manager of the CPU. It does not process data itself.

- * **Functions:** Fetches instructions, decodes them, and generates **timing and control signals** to direct the ALU, memory, and I/O devices.

4. Internal CPU Organization

The Registers, ALU, and CU are connected via an **Internal CPU Bus**. This localized data pathway ensures high-speed data transfer between CPU components without having to use the slower external system bus.

★ **Must-Write Points (for 10 marks)**

1. CPU has 3 main components: **Registers, ALU, and Control Unit (CU)**.
2. **Registers**: Extremely fast, on-chip storage for temporary data and addresses.
3. **PC (Program Counter)** stores next instruction address; **IR** holds current instruction.
4. **MAR/MBR** handle memory addresses/data; **Accumulator** stores math results.
5. **ALU**: Performs all Arithmetic (add, sub) and Logical (AND, OR) operations.
6. **Control Unit (CU)**: Decodes instructions and issues control signals to direct hardware.
7. Components communicate over high-speed **Internal CPU Buses**.

⚡ **Quick Recall**

CPU Org → Registers (PC/IR/MAR/MBR/ACC) → ALU (Math + Logic) → CU (Manager/Signals) → Internal Bus connects them